$C^{(12)[1]}_{\phi l}\,\mathtt{i1\_i2\_} \rightarrow$

$$
\begin{aligned}
&\frac{1}{16\pi^2}\Big(-\frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \\
&\quad \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right]\delta_{i1i2} + \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right]\delta_{i1i2} + \\
&\quad \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} + \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} - \\
&\quad \frac{3}{8}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} + \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} + \\
&\quad \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} + \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} - \\
&\quad \frac{5}{8}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} + \frac{1}{2}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} + \\
&\quad \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right]\delta_{i1i2} - \frac{1}{24}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right]\delta_{i1i2} - \\
&\quad \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \\
&\quad \frac{1}{4}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \frac{1}{2}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \\
&\quad \frac{1}{8} m_1 \tilde{\mu}\, g_1^4\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1]\,\delta_{i1i2} + \frac{1}{6} m_1 \tilde{\mu}\, g_1^4\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2} - \\
&\quad \frac{3}{8} m_2 \tilde{\mu}\, g_1^2 g_2^2\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2} + \frac{1}{2} m_2 \tilde{\mu}\, g_1^2 g_2^2\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} - \\
&\quad \frac{1}{8} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] - \frac{1}{16} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] + \\
&\quad \frac{1}{8} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] + \frac{1}{4} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] - \\
&\quad \frac{1}{4} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{1}{16} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{l}}^{i2}, m_{\tilde{e}}^{p}\right] - \\
&\quad \frac{3}{16} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{e}}^{p}\right] + \frac{1}{16} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] - \\
&\quad \frac{3}{16} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{e}}^{p}\right] - \frac{3}{16} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] - \\
&\quad \frac{1}{8} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] + \frac{1}{8} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] + \\
&\quad \frac{3}{8} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] - \frac{3}{8} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] + \\
&\quad \frac{1}{16}\tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \\
&\quad \frac{1}{32}\tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] + \\
&\quad \frac{1}{16} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] + \frac{1}{32} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] - \frac{3}{16}\tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] + \\
&\quad \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \frac{3}{16} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p} \\
&\qquad \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] - \frac{3}{32} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] - \\
&\quad \frac{1}{16}\tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{i2}\right] + \frac{1}{32}\tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} a_e^{i1p} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \frac{1}{32} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] - \\
&\quad \frac{3}{16}\tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{i2}\right] - \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} a_e^{i1p} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] + \frac{3}{32} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] + \\
&\quad \frac{1}{4}\tilde{\mu}\, \overline{y_e}^{rs}\, \overline{y_e}^{i2p} y_e^{i1s} a_e^{rp}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{r}, m_{\tilde{e}}^{p}, m_{\tilde{e}}^{s}, \tilde{\mu}\right] - \frac{1}{32}\tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} a_e^{i1p} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \\
&\quad \frac{1}{32} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i2}, m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] - \frac{3}{32} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i2}, m_2, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{1}{32}\tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} a_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] - \\
&\quad \frac{1}{32} m_1 \tilde{\mu}\, g_1^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] - \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} a_e^{i1p} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] + \frac{3}{32} m_2 \tilde{\mu}\, g_2^2\, \overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right]\Big)
\end{aligned}
$$